

CHAPTER 3

CELL DECOMPOSITION

Introduction

In this chapter we establish two important results in the subject of o-minimality: the monotonicity theorem (Section 1) and the cell decomposition theorem (Section 2). They are essential for everything that follows.

We work here with a fixed but arbitrary o-minimal structure $(R, <, \mathcal{S})$.

Instead of saying that a set $A \subseteq R^m$ belongs to $\mathcal{S}$ we will say that A is **definable**, as is usual in the literature on o-minimal structures. Similarly with maps.

For further conventions and notations, see Chapter 1, Sections 1, 2, 3.

§1. The monotonicity theorem and the finiteness lemma

(1.1) The monotonicity theorem describes definable one-variable functions.

(1.2) **MONOTONICITY THEOREM.** *Let $f: (a, b) \rightarrow R$ be a definable function on the interval (a, b) . Then there are points $a_1 < \cdots < a_k$ in (a, b) such that on each subinterval (a_j, a_{j+1}) , with $a_0 = a$, $a_{k+1} = b$, the function is either constant, or strictly monotone and continuous.*

(1.3) We derive this from the three lemmas below. In these lemmas we consider a definable function $f: I \rightarrow R$ on an interval I .

LEMMA 1. *There is a subinterval of I on which f is constant or injective.*

LEMMA 2. *If f is injective, then f is strictly monotone on a subinterval of I .*

LEMMA 3. *If f is strictly monotone, then f is continuous on a subinterval of I .*

(1.4) These lemmas imply the monotonicity theorem as follows:

Let

$$X := \{x \in (a, b) : \text{on some subinterval of } (a, b) \text{ containing } x \text{ the function } f \\ \text{is either constant, or strictly monotone and continuous}\}.$$

Now $(a, b) - X$ must be finite, since otherwise it would contain an interval I ; applying successively lemmas 1, 2, and 3 we can make I so small that f is either constant, or strictly monotone and continuous, on I . But then $I \subseteq X$, a contradiction.

Since $(a, b) - X$ is finite, we can reduce the proof of the theorem to the case that $(a, b) = X$, by replacing (a, b) by each of the finitely many intervals of which the open set X consists. In particular, we may assume that f is continuous. By splitting up (a, b) further we can reduce to one of the following three cases.

CASE 1. For all $x \in (a, b)$, f is constant on some neighborhood of x .

CASE 2. For all $x \in (a, b)$, f is strictly increasing on some neighborhood of x .

CASE 3. For all $x \in (a, b)$, f is strictly decreasing on some neighborhood of x .

Case 1. Take $x_0 \in (a, b)$ and put

$$s := \sup\{x : x_0 < x < b, f \text{ is constant on } [x_0, x]\}.$$

Then $s = b$, since $s < b$ implies that f is constant on some neighborhood of s , contradiction. From $s = b$ it follows that f is constant on $[x_0, b)$. Similarly, we prove that f is constant on $(a, x_0]$. Therefore f is constant on (a, b) .

Case 2. Take $x_0 \in (a, b)$ and put

$$s := \sup\{x : x_0 < x < b, f \text{ is strictly increasing on } [x_0, x]\}.$$

Then $s = b$, since $s < b$ leads to a contradiction as in case 1. Therefore f is strictly increasing on $[x_0, b)$. Similarly, f is strictly increasing on $(a, x_0]$. Hence f is strictly increasing on (a, b) .

Case 3. This is handled in the same way as case 2.

(1.5) We now prove the lemmas.

PROOF OF LEMMA 1. If some $y \in R$ had infinite preimage $f^{-1}(y)$, then this preimage would contain a subinterval of I and f would take the constant value y on that subinterval. So we may assume that each $y \in R$ has finite preimage. Then $f(I)$ is infinite, and so contains an interval J . Define an “inverse” $g: J \rightarrow I$ by

$$g(y) := \min\{x \in I : f(x) = y\}.$$

Since g is injective by definition, $g(J)$ is infinite, and hence $g(J)$ contains a subinterval of I , and f is necessarily injective on this subinterval. $\square$

PROOF OF LEMMA 2. Let us write $I = (a, b)$. We assume here that f is injective and have to show that f is strictly monotone on some subinterval of I . For each $x \in I$ the interval (a, x) is a disjoint union of two subsets,

$$(a, x) = \{y \in (a, x) : f(y) < f(x)\} \cup \{y \in (a, x) : f(y) > f(x)\},$$

so one of the parts contains an interval (c, x) , $a < c < x$. The interval (x, b) breaks up similarly. This shows that each $x \in I$ satisfies exactly one of the following four formulas:

$$\begin{aligned} \Phi_{++}(x) := \exists c_1, c_2 \in I \ [c_1 < x < c_2 \ \& \ \forall y \in (c_1, x) : f(y) > f(x) \\ \& \ \forall y \in (x, c_2) : f(y) > f(x)], \end{aligned}$$

$$\begin{aligned} \Phi_{+-}(x) := \exists c_1, c_2 \in I \ [c_1 < x < c_2 \ \& \ \forall y \in (c_1, x) : f(y) > f(x) \\ \& \ \forall y \in (x, c_2) : f(y) < f(x)], \end{aligned}$$

and $\Phi_{-+}(x)$ and $\Phi_{--}(x)$, which are defined similarly.

So I contains a subinterval all of whose points satisfy the same formula. Replacing I by this subinterval, and f by its restriction to that subinterval, we may assume that all points satisfy the same formula. This leads to four cases.

EASY CASE. $\Phi_{-+}(x)$ for all x in I .

For each x in I define $s(x) := \sup\{s \in (x, b) : f > f(x) \text{ on } (x, s]\}$. Then clearly $s(x) = b$, since $s(x) < b$ contradicts $\Phi_{-+}(s(x))$. Therefore f is strictly increasing on I .

The case that $\Phi_{+-}(x)$ for all x in I leads similarly to the conclusion that f is strictly decreasing on I .

DIFFICULT CASE. $\Phi_{++}(x)$ for all x in I .

Let $B := \{x \in I : \forall y \in I (y > x \text{ implies } f(y) > f(x))\}$. If B is infinite then B contains an interval, and on this interval f is strictly increasing, and we are done. So let us assume that B is finite. Passing to a subinterval to the right of all points of B we may assume

$$(*) \quad \forall x \in I \ \exists y \in I (y > x \ \& \ f(y) < f(x)).$$

Let $c \in I$. We claim that for all large enough y in I we have $f(y) < f(c)$. Otherwise, we would have $f(y) > f(c)$ for all large enough $y \in (c, b)$. Take then $d \in [c, b)$ minimal such that $\forall y (d < y < b \text{ implies } f(y) > f(c))$.

If $f(d) > f(c)$ then d would not be minimal since $\Phi_{++}(d)$. So $f(d) < f(c)$. But by $(*)$, there is e with $d < e < b$ and $f(e) < f(d)$, so $f(e) < f(c)$, contradiction. This proves the claim that $f(y) < f(c)$ for all large enough $y \in I$.

Define $y(c)$ as the least element of $[c, b]$ for which $f(y) < f(c)$ if $y(c) < y < b$. Note that $\Phi_{++}(c)$ gives $c < y(c)$ and $f(y(c)) < f(c)$. Minimality of $y(c)$ clearly implies that $y(c)$ satisfies the following formula $\Psi_{+-}(v)$:

$$\Psi_{+-}(v) := \exists v_1, v_2 \in I [v_1 < v < v_2 \ \& \ \forall z_1, z_2 (v_1 < z_1 < v < z_2 < v_2 \rightarrow f(z_1) > f(z_2))].$$

Since c was arbitrary we have shown $\forall c \in I \exists v \in I (v > c \ \& \ \Psi_{+-}(v))$.

Therefore $\Psi_{+-}(v)$ holds for all v in an interval of the form (d, b) , $d \in I$. Replacing I by this subinterval we may as well assume that $\Psi_{+-}(v)$ holds on all of I .

A completely similar argument shows that we can pass to a still smaller subinterval on which Ψ_{-+} (defined in the obvious way) holds. But this is a contradiction since we cannot simultaneously have $\Psi_{+-}(v)$ and $\Psi_{-+}(v)$. The case that $\Phi_{--}(x)$ holds for all x in I is completely similar to the case we just handled. This finishes the proof of lemma 2. $\square$

PROOF OF LEMMA 3. Let us assume that the strictly monotone function f is strictly increasing. (The case that f is strictly decreasing goes the same way.) Since $f(I)$ is infinite there is an interval $J \subseteq f(I)$. Take two points $r, s \in J$, $r < s$, and let c, d be their preimages: $f(c) = r$, $f(d) = s$, $c < d$. Clearly f defines an order preserving bijection of (c, d) onto (r, s) . But the topology is defined in terms of the order, hence f is continuous on (c, d) .

This finishes the proof of the lemmas and hence the proof of the monotonicity theorem is complete. $\square$

Let us mention two easy but important consequences.

(1.6) COROLLARY 1. *Let $f : (a, b) \rightarrow R$ be definable. Then for each $c \in (a, b)$ the limits $\lim_{x \uparrow c} f(x)$ and $\lim_{x \downarrow c} f(x)$ exist in R_∞ . Also the limits $\lim_{x \uparrow b} f(x)$ and $\lim_{x \downarrow a} f(x)$ exist in R_∞ .*

COROLLARY 2. *Let $f : [a, b] \rightarrow R$ be continuous and definable. Then f takes a maximum and a minimum value on $[a, b]$.*

Here is the other basic result of this section.

(1.7) FINITENESS LEMMA. *Let $A \subseteq R^2$ be definable and suppose that for each $x \in R$ the fiber $A_x := \{y \in R : (x, y) \in A\}$ is finite. Then there is $N \in \mathbb{N}$ such that $|A_x| \leq N$ for all $x \in R$.*

PROOF. A point $(a, b) \in R^2$ will be called **normal** if there is a box $I \times J$ around (a, b) such that

- either $(I \times J) \cap A = \emptyset$ (hence $(a, b) \notin A$),
 or $(a, b) \in A$ and $(I \times J) \cap A = \Gamma(f)$ for some continuous function $f: I \rightarrow R$.

(Note that in the latter case f is necessarily unique and definable.) Also, a point $(a, -\infty) \in R \times R_\infty$ is called **normal** if there is a box $I \times J$ disjoint from A such that $a \in I$ and $J = (-\infty, b)$ for some b . Finally, $(a, +\infty) \in R \times R_\infty$ is called **normal** if there is a box $I \times J$ disjoint from A with $a \in I$ and $J = (b, +\infty)$ for some b . Note

(1) The sets

$$\begin{aligned} &\{(a, b) \in R^2 : (a, b) \text{ is normal}\}, \\ &\{a \in R : (a, -\infty) \text{ is normal}\}, \\ &\{a \in R : (a, +\infty) \text{ is normal}\}, \end{aligned}$$

are definable.

Next we define functions $f_1, f_2, \dots, f_n, \dots$ by

$$\begin{aligned} \text{dom}(f_n) &:= \{x \in R : |A_x| \geq n\}, \\ \text{and } f_n(x) &:= n^{\text{th}} \text{ element of } A_x. \end{aligned}$$

Note that f_n is definable. (Its domain may of course be empty.)

Let $a \in R$ and take $n \geq 0$ maximal such that $f_1, f_2, \dots, f_n$ are defined and continuous on an interval containing a . We call the point a **good** or **bad**, according to whether

$$\begin{aligned} a \notin \text{cl}(\text{dom}(f_{n+1})) &\quad - \text{“good”}, \\ a \in \text{cl}(\text{dom}(f_{n+1})) &\quad - \text{“bad”}. \end{aligned}$$

Let $\mathcal{G}$ be the set of good points and $\mathcal{B}$ the set of bad points. Note that if $a \in \mathcal{G}$ then (with n as above) the domain of f_{n+1} is disjoint from an entire interval around a on which $f_1, f_2, \dots, f_n$ are defined and continuous. This shows that for $a \in \mathcal{G}$ we have

(2) $|A_x|$ is constant on an interval around a .

(3) (a, b) is normal for all $b \in R_\infty$.

The key point of the proof is to show that $\mathcal{B}$ and $\mathcal{G}$ are definable sets. (This is not clear from their definitions which involve a parameter $n \in \mathbb{N}$ depending on a .) To this end we shall show

(4) If $a \in \mathcal{B}$ then there is a least $b \in R_\infty$ such that (a, b) is not normal.

To see this, let us introduce for $a \in \mathcal{B}$ the elements $\lambda(a, -)$, $\lambda(a, 0)$, $\lambda(a, +)$ of R_∞ , where n has the same meaning as before:

$$\begin{aligned}\lambda(a, -) &:= \lim_{x \uparrow a} f_{n+1}(x) \text{ if } f_{n+1} \text{ is defined on some interval } (t, a), \\ &:= +\infty \text{ otherwise,} \\ \lambda(a, 0) &:= f_{n+1}(a) \text{ if } a \in \text{dom}(f_{n+1}), \\ &:= +\infty \text{ otherwise,} \\ \lambda(a, +) &:= \lim_{x \downarrow a} f_{n+1}(x) \text{ if } f_{n+1} \text{ is defined on some interval } (a, t), \\ &:= +\infty \text{ otherwise.}\end{aligned}$$

Now let $\beta(a) := \min\{\lambda(a, -), \lambda(a, 0), \lambda(a, +)\}$. Then by checking the various possibilities it is easy to see that $\beta(a)$ is the least $b \in R_\infty$ such that (a, b) is not normal. This proves claim (4), and together with (1) and (3) it implies that $\mathcal{B}$ and $\mathcal{G}$ are definable sets.

The rest of the proof is straightforward. Suppose first that $\mathcal{B}$ is finite, say

$$\mathcal{B} = \{a_1, \dots, a_k\}, \text{ with } -\infty = a_0 < a_1 < \dots < a_k < a_{k+1} = +\infty.$$

We claim that $|A_x|$ is constant on each interval (a_i, a_{i+1}) ; just take any point a in this interval, and let $n = |A_a|$. Then by (2) the set $\{x \in (a_i, a_{i+1}) : |A_x| = n\}$ is open, and for the same reason the set $\{x \in (a_i, a_{i+1}) : |A_x| \neq n\}$ is open. Since both sets are definable the latter set must be empty.

Suppose now that $\mathcal{B}$ is not finite. We shall derive a contradiction from this assumption, and that will finish the proof. Recall that $\beta(a)$ is, for $a \in \mathcal{B}$, the least $b \in R_\infty$ such that (a, b) is not normal. Define the sets

$$\begin{aligned}\mathcal{B}_- &:= \{a \in \mathcal{B} : \exists y (y < \beta(a) \ \& \ (a, y) \in A)\}, \\ \mathcal{B}_+ &:= \{a \in \mathcal{B} : \exists y (y > \beta(a) \ \& \ (a, y) \in A)\},\end{aligned}$$

and the functions $\beta_- : \mathcal{B}_- \rightarrow R$ and $\beta_+ : \mathcal{B}_+ \rightarrow R$ by

$$\begin{aligned}\beta_-(a) &:= \max\{y : y < \beta(a) \ \& \ (a, y) \in A\}, \\ \beta_+(a) &:= \min\{y : y > \beta(a) \ \& \ (a, y) \in A\}.\end{aligned}$$

Since $\mathcal{B}$ is infinite by assumption, one of the sets $\mathcal{B}_- \cap \mathcal{B}_+$, $\mathcal{B}_- - \mathcal{B}_+$, $\mathcal{B}_+ - \mathcal{B}_-$, $\mathcal{B} - (\mathcal{B}_- \cup \mathcal{B}_+)$ is infinite, and each of these four cases leads to a contradiction. We only show this in the case that $\mathcal{B}_- \cap \mathcal{B}_+$ is infinite. (The other cases are similar.) Since β_- , β , and β_+ are definable functions, there is by the monotonicity theorem an interval $I \subseteq \mathcal{B}_- \cap \mathcal{B}_+$ on which each of the functions β_- , β , β_+ is continuous. Note that $\beta_- < \beta < \beta_+$ on I . Now I splits into two subsets:

$$\begin{aligned}&\{x \in I : (x, \beta(x)) \in A\}, \\ &\text{and } \{x \in I : (x, \beta(x)) \notin A\},\end{aligned}$$

and one of these subsets contains an interval. Replacing I by this subinterval we may assume that either $\Gamma(\beta|I) \subseteq A$, or $\Gamma(\beta|I) \cap A = \emptyset$. In either case it is clear that $\Gamma(\beta|I)$ consists of normal points, since β_- , β , β_+ are continuous on I . Now we have a contradiction, since $(a, \beta(a))$ is never normal. $\square$

(1.8) We now combine the finiteness lemma with the monotonicity theorem in the following result to be used in the next section.

Let $A \subseteq R^2$ be definable such that A_x is finite for each $x \in R$. Then there are points $a_1 < \dots < a_k$ in R such that the intersection of A with each vertical strip $(a_i, a_{i+1}) \times R$ has the form $\Gamma(f_{i1}) \cup \dots \cup \Gamma(f_{in(i)})$ for certain definable continuous functions $f_{ij}: (a_i, a_{i+1}) \rightarrow R$ with $f_{i1}(x) < \dots < f_{in(i)}(x)$ for $x \in (a_i, a_{i+1})$. (Here we have set $a_0 := -\infty$, $a_{k+1} := +\infty$.)

(1.9) EXERCISES.

1. Suppose the function $f: (a, b) \rightarrow R$ on the interval (a, b) is definable. Show there exist elements $a_1, \dots, a_k$ with the property of the monotonicity theorem such that $a_1, \dots, a_k$ are definable in the model-theoretic structure $(R, <, \Gamma(f))$.

2. Let I and J be intervals and $f: I \rightarrow R$ and $g: J \rightarrow R$ strictly monotone definable functions such that $f(I) \subseteq g(J)$ and $\lim_{s \rightarrow r(I)} f(s) = \lim_{t \rightarrow r(J)} g(t)$ in R_∞ , where $r(I)$ and $r(J)$ are the right endpoints of the intervals I and J in R_∞ . Show that near these right endpoints f and g are reparametrizations of each other, that is, there are subintervals I' of I and J' of J , with $r(I) = r(I')$, $r(J) = r(J')$ and a strictly increasing definable bijection $h: I' \rightarrow J'$ such that $f(s) = g(h(s))$ for all $s \in I'$. (This will be used in Chapter 6, (4.3).)

§2. The cell decomposition theorem

(2.1) In this section we show that a definable subset of R^m splits into finitely many cells—definable sets of an especially simple form—and that each definable function on a subset of R^m is “cellwise” continuous. For $m = 1$ this reduces for sets to the definition of “o-minimal” and for functions to the monotonicity theorem of the last section.

(2.2) For each definable set X in R^m we put

$$\begin{aligned} C(X) &:= \{f: X \rightarrow R : f \text{ is definable and continuous}\}, \\ C_\infty(X) &:= C(X) \cup \{-\infty, +\infty\}, \end{aligned}$$

where we regard $-\infty$ and $+\infty$ as constant functions on X .

For f, g in $C_\infty(X)$ we write $f < g$ if $f(x) < g(x)$ for all $x \in X$, and in this case we put

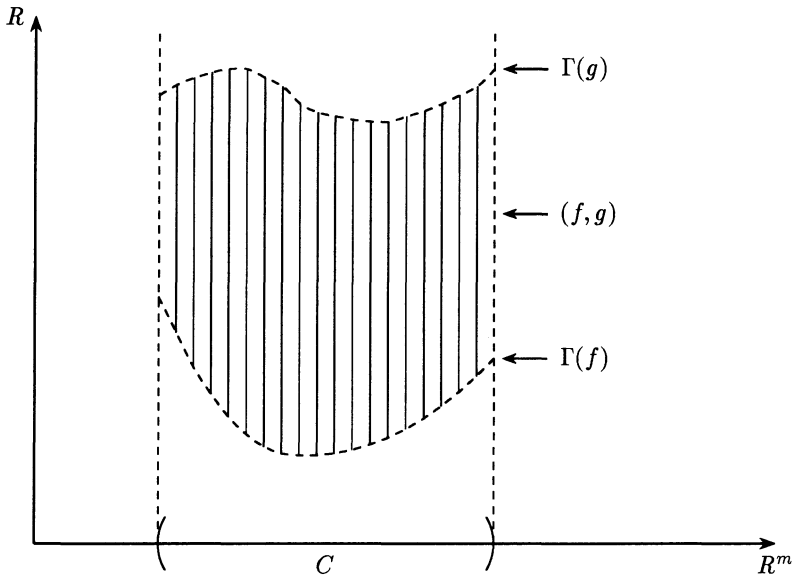
$$(f, g)_X := \{(x, r) \in X \times R : f(x) < r < g(x)\}.$$

So $(f, g)_X$ is a definable subset of R^{m+1} .

Usually we write just (f, g) instead of $(f, g)_X$ when X is clear from the context.

(2.3) DEFINITION. Let $(i_1, \dots, i_m)$ be a sequence of zeros and ones of length m . An $(i_1, \dots, i_m)$ -**cell** is a definable subset of R^m obtained by induction on m as follows:

- (i) a (0)-cell is a one-element set $\{r\} \subseteq R$ (a “point”), a (1)-cell is an interval $(a, b) \subseteq R$;
- (ii) suppose $(i_1, \dots, i_m)$ -cells are already defined; then an $(i_1, \dots, i_m, 0)$ -cell is the graph $\Gamma(f)$ of a function $f \in C(X)$, where X is an $(i_1, \dots, i_m)$ -cell; further, an $(i_1, \dots, i_m, 1)$ -cell is a set $(f, g)_X$ where X is an $(i_1, \dots, i_m)$ -cell and $f, g \in C_\infty(X)$, $f < g$. (See figure.)



So a (0,0)-cell is a “point” $\{(r, s)\} \subseteq R^2$, a (0,1)-cell is an “interval” on a vertical line $\{a\} \times R$, and a (1,0)-cell is the graph of a continuous definable function defined on an interval. Note that a box in R^m is a $(1, \dots, 1)$ -cell.

(2.4) TERMINOLOGY. A **cell** in R^m is an $(i_1, \dots, i_m)$ -cell, for some (necessarily unique) sequence $(i_1, \dots, i_m)$. Since the $(1, \dots, 1)$ -cells are exactly the cells which are open in their ambient space R^m we call these **open cells**.

(2.5) The non-open cells are “thin”:

The union of finitely many non-open cells in R^m has empty interior. This is easily

checked and we shall use this fact later on without further mention. Here is another topological fact:

Each cell is locally closed, i.e. open in its closure.

To see this, let $C \subseteq R^{m+1}$ be a cell. Put $B := \pi(C) \subseteq R^m$ and assume inductively that the cell B is open in its closure $\text{cl}(B)$, so that $\text{cl}(B) - B$ is a closed set. If $C = \Gamma(f)$ with $f: B \rightarrow R$ a definable continuous function, then $\text{cl}(C) - C$ is contained in $(\text{cl}(B) - B) \times R$, hence C is open in the closed set $C \cup ((\text{cl}(B) - B) \times R)$. If $C = (f, g)$ with $f, g: B \rightarrow R$ definable continuous functions on B , $f < g$, then one easily verifies that $\text{cl}(C) - C$ is contained in $\Gamma(f) \cup \Gamma(g) \cup ((\text{cl}(B) - B) \times R)$ and that C is open in the closed set $C \cup \Gamma(f) \cup \Gamma(g) \cup ((\text{cl}(B) - B) \times R)$. (Draw a picture.) The other cases are done similarly.

(2.6) CONVENTION. We also consider the one-point space R^0 as a cell, more precisely, as a $(\)$ -cell, where $(\)$ is the sequence of length 0. (It is an open cell; there are no other $(\)$ -cells.)

(In this way clause (i) in (2.3) appears as the case $m = 0$ of clause (ii). It allows us to start inductions with $m = 0$.)

(2.7) Each cell is homeomorphic under a coordinate projection to an open cell. We now make this explicit. Let $i = (i_1, \dots, i_m)$ be a sequence of zeros and ones.

Define $p_i: R^m \rightarrow R^k$ as follows: let $\lambda(1) < \dots < \lambda(k)$ be the indices $\lambda \in \{1, \dots, m\}$ for which $i_\lambda = 1$, so that $k = i_1 + \dots + i_m$; then

$$p_i(x_1, \dots, x_m) := (x_{\lambda(1)}, \dots, x_{\lambda(k)}).$$

It is easy to show by induction on m that p_i maps each i -cell A homeomorphically onto an open cell $p_i(A)$ in R^k . We denote $p_i(A)$ also by $p(A)$ and the homeomorphism $p_i|_A: A \rightarrow p(A)$ by p_A . Clearly $p_A = \text{id}_A$ if A is an open cell.

(2.8) If A is a cell in R^{m+1} then $\pi(A)$ is a cell in R^m , where $\pi: R^{m+1} \rightarrow R^m$ is the projection on the first m coordinates. Here is a simple application of this fact.

(2.9) PROPOSITION. *Each cell is definably connected.*

PROOF. For intervals and points this is stated in Chapter 1, (3.6). If A is a cell in R^{m+1} , then we assume inductively that the cell $\pi(A)$ in R^m is definably connected and use the fact that each fiber $\pi^{-1}(x) \cap A$ is definably connected. $\square$

(2.10) DEFINITION. A **decomposition** of R^m is a special kind of partition of R^m into finitely many cells. The definition is by induction on m :

(i) a decomposition of $R^1 = R$ is a collection

$$\{(-\infty, a_1), (a_1, a_2), \dots, (a_k, +\infty), \{a_1\}, \dots, \{a_k\}\}$$

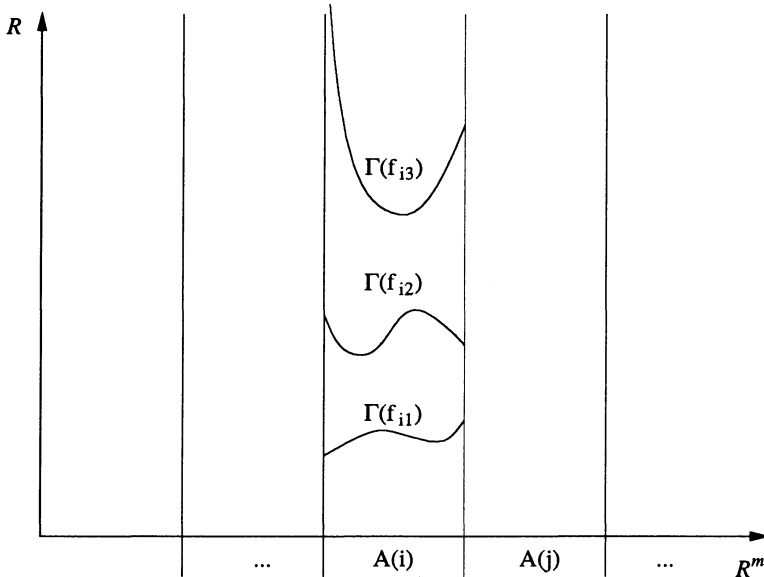
- where $a_1 < \dots < a_k$ are points in R ;
- (ii) a decomposition of R^{m+1} is a finite partition of R^{m+1} into cells A such that the set of projections $\pi(A)$ is a decomposition of R^m . (Here $\pi: R^{m+1} \rightarrow R^m$ is the usual projection map.)

Let $\mathcal{D} = \{A(1), \dots, A(k)\}$ be a decomposition of R^m , $A(i) \neq A(j)$ if $i \neq j$, and let for each $i \in \{1, \dots, k\}$ functions $f_{i1} < \dots < f_{in(i)}$ in $C(A_i)$ be given.

Then

$$\mathcal{D}_i := \{(-\infty, f_{i1}), (f_{i1}, f_{i2}), \dots, (f_{in(i)}, +\infty), \Gamma(f_{i1}), \dots, \Gamma(f_{in(i)})\}$$

is a partition of $A(i) \times R$ and one easily checks that $\mathcal{D}^* := \mathcal{D}_1 \cup \dots \cup \mathcal{D}_k$ is a decomposition of R^{m+1} , and that every decomposition of R^{m+1} arises in this way from a decomposition $\mathcal{D}$ of R^m . We write $\mathcal{D} = \pi(\mathcal{D}^*)$. (See figure.)



A decomposition $\mathcal{D}$ of R^m is said to **partition** a set $S \subseteq R^m$ if each cell in $\mathcal{D}$ is either part of S or disjoint from S , in other words, if S is a union of cells in $\mathcal{D}$. We are now ready to state the main result of this chapter.

(2.11) CELL DECOMPOSITION THEOREM.

- (I_m) Given any definable sets $A_1, \dots, A_k \subseteq R^m$ there is a decomposition of R^m partitioning each of $A_1, \dots, A_k$.
- (II_m) For each definable function $f: A \rightarrow R$, $A \subseteq R^m$, there is a decomposition $\mathcal{D}$ of R^m partitioning A such that the restriction $f|_B: B \rightarrow R$ to each cell $B \in \mathcal{D}$ with $B \subseteq A$ is continuous.

(2.12) The proof is by induction on m . Note that (I_1) holds by o-minimality, and that (II_1) follows easily from the monotonicity theorem.

We now assume that $(I_1), \dots, (I_m)$ and $(II_1), \dots, (II_m)$ hold, and shall derive first (I_{m+1}) and then (II_{m+1}) .

The proof is lengthy. The first step is to generalize the finiteness lemma of the previous section. Call a set $Y \subseteq R^{m+1}$ **finite over** R^m if for each $x \in R^m$ the fiber $Y_x := \{r \in R : (x, r) \in Y\}$ is finite; call Y **uniformly finite over** R^m if there is $N \in \mathbb{N}$ such that $|Y_x| \leq N$ for all $x \in R^m$. We shall use the finiteness lemma and the inductive assumption to prove first

(2.13) **LEMMA (UNIFORM FINITENESS PROPERTY).** *Suppose the definable subset Y of R^{m+1} is finite over R^m . Then Y is uniformly finite over R^m .*

PROOF. A box $B \subseteq R^m$ will be called **Y -good** if for each point $(x, r) \in Y$ with $x \in B$ there is an interval I around r such that $Y \cap (B \times I) = \Gamma(f)$ for some continuous function $f: B \rightarrow R$. (Note: this f is then uniquely determined by Y , B , and I , and is definable.)

Claim 1. Suppose the box $B \subseteq R^m$ is Y -good; then there are continuous definable functions $f_1 < \dots < f_k$ in $C(B)$ such that $Y \cap (B \times R) = \Gamma(f_1) \cup \dots \cup \Gamma(f_k)$.

To see this, let us fix $x \in B$ and write $Y_x = \{r_1, \dots, r_k\}$ with $r_1 < \dots < r_k$. Take intervals $I_1, \dots, I_k$ around $r_1, \dots, r_k$ respectively, and continuous functions $f_1, \dots, f_k: B \rightarrow R$ such that $Y \cap (B \times I_j) = \Gamma(f_j)$, $j = 1, \dots, k$.

Subclaim a. $f_1 < \dots < f_k$.

Let us prove only $f_1 < f_2$, the other inequalities following in the same way. Suppose there is a point $p \in B$ with $f_1(p) = f_2(p)$. So $f_2(p) \in I_1$, and by continuity of f_2 there is a neighborhood $U \subseteq B$ of p such that $f_2(U) \subseteq I_1$. Since $Y \cap (U \times I_1) = \Gamma(f_1|U)$ and $\Gamma(f_2|U) \subseteq Y \cap (U \times I_1)$ it follows that $f_1|U = f_2|U$. This argument shows that the set $\{p \in B : f_1(p) = f_2(p)\}$ is open. Since $\{p \in B : f_1(p) < f_2(p)\}$ and $\{p \in B : f_1(p) > f_2(p)\}$ are also open and B is definably connected, by (2.9), while $f_1(x) = r_1 < r_2 = f_2(x)$, it follows that $f_1 < f_2$.

Subclaim b. $Y \cap (B \times R) = \Gamma(f_1) \cup \dots \cup \Gamma(f_k)$.

Take any point $(a, s) \in Y \cap (B \times R)$ and let $f: B \rightarrow R$ be a continuous definable function such that $f(a) = s$ and $\Gamma(f) \subseteq Y$. Since $(x, f(x)) \in Y$ it follows that $f(x) = r_i = f_i(x)$ for some $i \in \{1, \dots, k\}$. As in the proof of subclaim a we obtain from this $f = f_i$. This finishes the proof of claim 1.

A point $x \in R^m$ will be called **Y -good** if x belongs to a Y -good box. Note that the set of Y -good points is definable.

Claim 2. If $A \subseteq R^m$ is a definably connected set and all points of A are Y -good,

then there are continuous functions $f_1 < \dots < f_k$ in $C(A)$ such that

$$Y \cap (A \times R) = \Gamma(f_1) \cup \dots \cup \Gamma(f_k).$$

To prove this we choose a point $x \in A$, if A is nonempty, and let $k = |Y_x|$. By claim 1 the set $\{a \in A : |Y_a| = k\}$ is open and closed in A , hence $|Y_a| = k$ for all $a \in A$. Again, by claim 1, it is clear that $f_1, \dots, f_k$ are continuous.

Claim 3. Each open cell in R^m contains a Y -good point.

It is of course enough to show that each box B in R^m contains a Y -good point. Write

$$B = B' \times (a, b), \quad B' \text{ a box in } R^{m-1}.$$

For each point $p \in B'$ consider the set

$$Y(p) := \{(r, s) \in R^2 : a < r < b \text{ \& } (p, r, s) \in Y\},$$

which is finite over R . Now we apply (1.8) to $A = Y(p)$ and conclude that the set $\{r \in R : r \text{ is not } Y(p)\text{-good}\}$ is finite. Therefore, the definable set

$$\text{Bad}(Y) := \{(p, r) \in B : r \text{ is not } Y(p)\text{-good}\}$$

has no interior point. By the inductive assumption (I_m) there is a decomposition of R^m which partitions B and $\text{Bad}(Y)$. Take an open cell C of this partition such that $C \subseteq B$. Then $C \cap \text{Bad}(Y) = \emptyset$, so if we replace B by a box contained in C we have reduced to the case that $\text{Bad}(Y) = \emptyset$, i.e., for each $p \in B'$ we can apply claim 2 (with $Y(p) \subseteq R^2$ instead of Y) to find a number $k(p) \in \mathbb{N}$ such that $|Y_x| = k(p)$ for each point $x = (p, r) \in B$. Next we have to show that there is a finite bound on the numbers $k(p)$, $p \in B'$.

To this end, we choose an $r \in (a, b)$ and consider the set

$$Y^r := \{(p, s) : (p, r, s) \in Y\} \subseteq R^m.$$

Since Y is finite over R^m , the set Y^r is finite over R^{m-1} , so by the inductive assumption Y^r is uniformly finite over R^{m-1} , i.e. there is $N \in \mathbb{N}$ such that for each $p \in B'$: $|\{s \in R : (p, s) \in Y^r\}| \leq N$, that is, $|Y_{(p,r)}| \leq N$ for all $p \in B'$. Hence $k(p) \leq N$ for all $p \in B'$. Thus $|Y_x| \leq N$ for all $x \in B$.

For each $i \in \{0, \dots, N\}$ let $B_i := \{x \in B : |Y_x| = i\}$, and define the functions $f_{i1}, \dots, f_{ii}$ on B_i by $f_{i1}(x) < \dots < f_{ii}(x)$, and $Y_x = \{f_{i1}(x), \dots, f_{ii}(x)\}$. Applying the inductive assumption (II_m) to each f_{ij} separately, and then using (I_m) to find a common refinement of the decompositions obtained via (II_m) , we get a decomposition $\mathcal{D}$ of R^m partitioning each of the sets B_i , such that for each $A \in \mathcal{D}$, if $A \subseteq B_i$, then $f_{ij}|_A$ is continuous, $j = 0, \dots, i$. Since B is open and is partitioned by $\mathcal{D}$, there is an open cell $A \in \mathcal{D}$ with $A \subseteq B$. Now $B = \bigcup_i B_i$, so $A \subseteq B_i$ for some i , therefore the functions $f_{i1}, \dots, f_{ij}$ are continuous on A . Hence each point of A is Y -good. Since $A \subseteq B$ this establishes claim 3.

The proof of the lemma now proceeds as follows. Take a decomposition $\mathcal{D}$ of R^m partitioning the set of Y -good points. Let $A \in \mathcal{D}$. If A is open, then by claim 3 the cell A contains a Y -good point, so all points of A are Y -good. By claim 2 there is then a number $N_A \in \mathbb{N}$ such that $|Y_x| \leq N_A$ for all $x \in A$. By an easy exercise using the definable homeomorphism p_A (see (2.7)) such a number N_A also exists for the non-open cells $A \in \mathcal{D}$. Now take $N := \max\{N_A : A \in \mathcal{D}\}$. Then $|Y_x| \leq N$ for all x in R^m . $\square$

(2.14) Before we proceed with the proof of (I_{m+1}) we recall from Chapter 1, (3.3) that a definable set $S \subseteq R$ has finite boundary $\text{bd}(S)$, and that the interval between two successive boundary points is either part of S or disjoint from S .

For a definable set $A \subseteq R^{m+1}$ we put

$$\text{bd}_m(A) := \{(x, r) \in R^{m+1} : r \in \text{bd}(A_x)\},$$

and we note that $\text{bd}_m(A)$ is a definable set which is finite over R^m , so that we can apply the uniform finiteness property. This is the idea of the proof below.

(2.15) PROOF OF (I_{m+1}) . Let $A_1, \dots, A_k$ be definable subsets of R^{m+1} . Put

$$Y := \text{bd}_m(A_1) \cup \dots \cup \text{bd}_m(A_k).$$

Then $Y \subseteq R^{m+1}$ is definable and finite over R^m , so there is $M \in \mathbb{N}$ such that $|Y_x| \leq M$ for all x in R^m . For each $i \in \{0, \dots, M\}$ let $B_i := \{x \in R^m : |Y_x| = i\}$, and define functions $f_{i1}, f_{i2}, \dots, f_{ii}$ on B_i by

$$Y_x = \{f_{i1}(x), \dots, f_{ii}(x)\}, \quad f_{i1}(x) < \dots < f_{ii}(x).$$

Further, put $f_{i0} := -\infty$, $f_{ii+1} := +\infty$ (functions on B_i). Finally we define for each $\lambda \in \{1, \dots, k\}$, $i \in \{0, \dots, M\}$ and $1 \leq j \leq i$

$$C_{\lambda ij} := \{x \in B_i : f_{ij}(x) \in (A_\lambda)_x\},$$

and for each $\lambda \in \{1, \dots, k\}$, $i \in \{0, \dots, M\}$ and $0 \leq j \leq i$

$$D_{\lambda ij} := \{x \in B_i : (f_{ij}(x), f_{i,j+1}(x)) \subseteq (A_\lambda)_x\}.$$

We now take a decomposition $\mathcal{D}$ of R^m which partitions each set B_i , each set $C_{\lambda ij}$, each set $D_{\lambda ij}$, and which has moreover the following property:

if $E \in \mathcal{D}$ is contained in B_i , then $f_{i1}|_E, \dots, f_{ii}|_E$ are continuous functions.

(Such a decomposition exists by the inductive assumptions (I_m) and (II_m) .)

For each cell $E \in \mathcal{D}$ we let $\mathcal{D}_E$ be the following partition of $E \times R$:

$$\mathcal{D}_E := \{(f_{i0}|_E, f_{i1}|_E), \dots, (f_{ii}|_E, f_{ii+1}|_E), \Gamma(f_{i1}|_E), \dots, \Gamma(f_{ii}|_E)\},$$

where $i \in \{0, \dots, M\}$ is such that $E \subseteq B_i$. Then $\mathcal{D}^* := \bigcup \{\mathcal{D}_E : E \in \mathcal{D}\}$ is a decomposition of R^{m+1} which partitions each set $A_1, \dots, A_k$. $\square$

The proof of (II_{m+1}) will be based on the following elementary lemma.

(2.16) LEMMA. Let X be a topological space, $(R_1, <)$, $(R_2, <)$ dense linear orderings without endpoints and $f : X \times R_1 \rightarrow R_2$ a function such that for each $(x, r) \in X \times R_1$

- (i) $f(x, \cdot) : R_1 \rightarrow R_2$ is continuous and monotone on R_1 ,
- (ii) $f(\cdot, r) : X \rightarrow R_2$ is continuous at x .

Then f is continuous.

PROOF. Let $(x, r) \in X \times R_1$ and $f(x, r) \in J$, where J is an interval in R_2 . We shall find a neighborhood U of x and an interval I around r such that $f(U \times I) \subseteq J$. By (i) there are r_-, r_+ in R_1 such that $r_- < r < r_+$ and $f(x, r_-), f(x, r_+) \in J$. Now use (ii) to get a neighborhood U of x such that $f(U \times \{r_-\}) \subseteq J$ and $f(U \times \{r_+\}) \subseteq J$. We claim that then $f(U \times I) \subseteq J$ for $I = (r_-, r_+)$.

Let $x' \in U$ and $r_- < r' < r_+$. Assume $f(x', \cdot)$ is increasing. (The case that $f(x', \cdot)$ is decreasing goes the same way.) Then $f(x', r_-) \leq f(x', r') \leq f(x', r_+)$ and $f(x', r_-), f(x', r_+)$ are both in J , hence $f(x', r')$ is in J . $\square$

(2.17) PROOF OF (II_{m+1}) . Let $f : A \rightarrow R$ be a definable function on a definable set $A \subseteq R^{m+1}$. We have to show that f is “cellwise” continuous. Because of (I_{m+1}) it suffices to show

- (*) $\left\{ \begin{array}{l} A \text{ can be partitioned into finitely many definable sets } A_1, \dots, A_k \\ \text{such that } f|_{A_i} : A_i \rightarrow R \text{ is continuous for } i = 1, \dots, k. \end{array} \right.$

Again because of (I_{m+1}) the set A allows a partition into finitely many cells. So in order to prove (*) we may assume that A is already a cell. If the cell A is not open in R^{m+1} , we use the definable homeomorphism $p_A : A \rightarrow p(A)$. Since $p(A) \subseteq R^n$ for some $n \leq m$, it follows from the inductive assumption (II_n) that the set $p(A)$ can be partitioned into definable sets $B_1, \dots, B_k$ such that $(f \circ p_A^{-1})|_{B_j}$ is continuous for each j . Hence A is partitioned into $p_A^{-1}(B_1), \dots, p_A^{-1}(B_k)$, and the restriction of f to each of these sets is continuous.

This establishes (*) for A non-open. Suppose now that A is an open cell. Call f **well-behaved** at a point $(p, r) \in A$ if $p \in C$ for some box $C \subseteq R^m$ and $a < r < b$ for some a, b in R such that

- (i) $C \times (a, b)$ is contained in A ,
- (ii) for all $x \in C$ the function $f(x, \cdot)$ is continuous and monotone on (a, b) ,
- (iii) the function $f(\cdot, r)$ is continuous at p .

Let A^* be the set of all points of A at which f is well-behaved. Note that A^* is definable.

Claim. A^* is dense in A .

To prove this it suffices to show that, given any box $B \subseteq R^m$ and $-\infty < a < c < +\infty$ such that $B \times (a, c)$ is contained in A , the box $B \times (a, c)$ intersects A^* . Now

there is by the monotonicity theorem for each $x \in B$ a largest $\lambda(x) \in (a, c]$ such that the one-variable function $f(x, \cdot)$ is continuous and monotone on $(a, \lambda(x))$. Since $\lambda: B \rightarrow R$ is definable, there is by (II_m) a box $C \subseteq B$ on which λ is continuous. Taking C small enough we may assume that $b \leq \lambda(x)$ for all $x \in C$ and a fixed $b \in (a, c)$. Choose any element r in (a, b) . The function $f(\cdot, r): C \rightarrow R$ is continuous on some smaller box, by (II_m). Replacing C by this smaller box, we see that f is well-behaved at each point (p, r) with p in C . This establishes the claim.

Now we take a decomposition $\mathcal{D}$ of R^{m+1} that partitions both A and A^* . (Such a decomposition exists by (I_{m+1}).) Let $D \in \mathcal{D}$ be any open cell contained in A .

We need only show that f is continuous on D . From $D \subseteq A$ we obtain $D \subseteq A^*$, since D intersects A^* by the claim, and $\mathcal{D}$ partitions A^* . In particular for each point (p, r) in D the function $f(\cdot, r)$ is continuous at p . Therefore D is the union of boxes $C \times (a, b)$ satisfying the conditions (i), (ii), and (iii) above, for each point $p \in C$, $a < r < b$. By lemma (2.16) the function f is continuous on each such box, hence f is continuous on D . This concludes the proof of the cell decomposition theorem. $\square$

We finish this section with an application of cell decomposition to definably connected components, and some exercises.

A definably connected component of a nonempty definable set $X \subseteq R^m$ is by definition a maximal definably connected subset of X .

(2.18) PROPOSITION. *Let $X \subseteq R^m$ be a nonempty definable set. Then X has only finitely many definably connected components. They are open and closed in X and form a finite partition of X .*

PROOF. Let $\{C_1, \dots, C_k\}$ be a partition of X into k disjoint cells. For each nonempty set of indices $I \subseteq \{1, \dots, k\}$, put $C_I := \bigcup_{i \in I} C_i$. Among the $2^k - 1$ sets C_I , let C' be maximal with respect to being definably connected. We claim

If a set $Y \subseteq X$ is definably connected and $C' \cap Y \neq \emptyset$, then $Y \subseteq C'$.

To see why, put $C_Y := \bigcup \{C_i : C_i \cap Y \neq \emptyset\}$. Since the C_i 's cover X we have $Y \subseteq C_Y$, so C_Y is the union of Y with certain cells that intersect Y . Hence C_Y is definably connected. But $C' \cap C_Y$ contains the nonempty set $C' \cap Y$, so $C' \cup C_Y$ is definably connected. By maximality of C' it follows that $C' \cup C_Y = C'$. Hence $Y \subseteq C_Y \subseteq C'$, which proves the claim. It follows in particular that C' is a definably connected component of X . It also follows that the sets of the form C' form a (finite) partition of X . Further the claim shows that the sets C' are the only definably connected components of X . Note that because the closure in X of a definably connected subset of X is also definably connected, the definably connected components of X are closed in X . Hence they are also open in X . $\square$

(2.19) EXERCISES.

In the first four exercises we strengthen the cell decomposition theorem in several ways. This requires a somewhat technical notion of regularity:

- (i) Call an open cell $C \subseteq R^m$ **regular** if for each $i \in \{1, \dots, m\}$ and each two points $x, y \in C$ that differ only in the i^{th} coordinate and each point $z \in R^m$ that differs from x and y only in the i^{th} coordinate, we have $x_i < z_i < y_i \Rightarrow z \in C$. (Example: boxes in R^m are regular.)
- (ii) Consider a definable function $f: C \rightarrow R$, where $C \subseteq R^m$ is a regular open cell. Call f **strictly increasing in the i^{th} coordinate** ($1 \leq i \leq m$) if for all points x, y in C that differ only in the i^{th} coordinate, with $x_i < y_i$, we have $f(x) < f(y)$; similarly we define the notions of “ f is **strictly decreasing in the i^{th} coordinate**” and “ f is **independent of the i^{th} coordinate**”, the latter meaning that $f(x) = f(y)$ whenever $x, y \in C$ differ only in the i^{th} coordinate. Finally, let us say that f is **regular** if f is continuous, and for each $i \in \{1, \dots, m\}$ the function f is either strictly increasing in the i^{th} coordinate, or strictly decreasing in the i^{th} coordinate, or independent of the i^{th} coordinate. (Which of these three cases takes holds may depend on i .)

1. Let $C \subseteq R^m$ be a regular open cell and $f: C \rightarrow R$ a regular definable function. Show that the open cells $(-\infty, f)$, $(f, +\infty)$ and $C \times R = (-\infty, +\infty)$ in R^{m+1} are regular. Show that if $g: C \rightarrow R$ is a second regular definable function with $f < g$, then the open cell (f, g) in R^{m+1} is regular.

2. Prove by induction on m the **regular cell decomposition theorem**:

- (I_m) For any definable sets $A_1, \dots, A_k \subseteq R^m$ there is a decomposition of R^m partitioning each A_i , all of whose open cells are regular.
- (II_m) For each definable function $f: A \rightarrow R$, $A \subseteq R^m$, there is a decomposition $\mathcal{D}$ of R^m partitioning A all of whose open cells are regular, and such that for each open cell $C \in \mathcal{D}$ with $C \subseteq A$ the restriction $f|_C$ is regular.

3. Let C be a cell in R^m , $D = (\alpha, \beta)_C$ a cell in R^{m+1} and $f: D \rightarrow R$ a definable function such that for all $x \in C$ the function $f(x, \cdot): (\alpha(x), \beta(x)) \rightarrow R$ is continuous. Show that C can be partitioned into cells $C_1, \dots, C_k$ such that, with $\alpha_i := \alpha|_{C_i}$, $\beta_i := \beta|_{C_i}$, each restriction $f|_{(\alpha_i, \beta_i)}: (\alpha_i, \beta_i) \rightarrow R$ is continuous.

4. Improve the cell decomposition theorem as follows:

- (I_m) If the sets $A_1, \dots, A_k \subseteq R^m$ are definable, then there is a decomposition of R^m partitioning each set A_i , all of whose cells are definable in the model-theoretic structure $(R, <, A_1, \dots, A_k)$.
- (II_m) Let the function $f: A \rightarrow R$, $A \subseteq R^m$, be definable. Then there is a decomposition $\mathcal{D}$ of R^m partitioning A , such that the restriction $f|_B$ to each cell $B \in \mathcal{D}$ with $B \subseteq A$ is continuous, and each cell in $\mathcal{D}$ is definable in the model-theoretic structure $(R, <, \Gamma(f))$.

5. Let $X_1, \dots, X_k \subseteq R^m$ be distinct nonempty definably connected sets and X their union. Define a graph with vertex set $\{X_1, \dots, X_k\}$ by putting an edge between X_i and X_j ($i \neq j$) if $X_i \cap \text{cl}(X_j) \neq \emptyset$ or $\text{cl}(X_i) \cap X_j \neq \emptyset$.

Show that if $X_{i(1)}, \dots, X_{i(r)}$ are the vertices of a connected component of this graph, then $X_{i(1)} \cup \dots \cup X_{i(r)}$ is a definably connected component of X , and that all definably connected components of X are of this form. (This gives a useful method to construct the definably connected components of X from a partition of X into finitely many cells.)

6. Suppose $\mathcal{S}'$ is an o-minimal structure on $(R, <)$ with $\mathcal{S} \subseteq \mathcal{S}'$, and let $X \subseteq R^m$ belong to $\mathcal{S}$, so X also belongs to $\mathcal{S}'$. Show that X is definably connected in the sense of $\mathcal{S}$ if and only if X is definably connected in the sense of $\mathcal{S}'$.

7. Suppose $\mathcal{S}$ is an o-minimal structure on the ordered set $(\mathbb{R}, <)$ of real numbers. Show that for a definable set $X \subseteq \mathbb{R}^m$ the following are equivalent:

- (a) X is definably connected;
- (b) X is connected in the usual topological sense.

REMARK. If $\mathcal{S}$ contains addition, then (a) and (b) above are also equivalent to “ X is definably path connected”, see Chapter 6, (3.2), and also Chapter 6, (1.15), exercise 8.

8. With the same hypothesis as in exercise 7, show that each definable set $X \subseteq \mathbb{R}^m$ is locally connected, that is, for each $x \in X$ and each open subset U of X containing x there is a connected open subset V of X containing x and contained in U .

§3. Definable families

(3.1) Let $S \subseteq R^{m+n} = R^m \times R^n$ be definable. For each $a \in R^m$ we put

$$S_a := \{x \in R^n : (a, x) \in S\}, \text{ a subset of } R^n.$$

We view S as describing the family of sets $(S_a)_{a \in R^m}$. Such a family is called a **definable family** (of subsets of R^n , with parameter space R^m). The sets S_a are also called the **fibers** of the family.

(3.2) EXAMPLE. Let $\mathcal{R} := (\mathbb{R}, <, +, \cdot)$, and consider the formula

$$(*) \quad ax^2 + bxy + cy^2 + dx + ey + f = 0.$$

This defines a relation $S \subseteq \mathbb{R}^6 \times \mathbb{R}^2$. For each point (a, b, c, d, e, f) in $\mathbb{R}^6$ the subset $S_{(a,b,c,d,e,f)}$ of $\mathbb{R}^2$ consists of the points (x, y) satisfying $(*)$. We know that such a set can be an ellipse, a parabola, a hyperbola, two intersecting lines, two parallel lines, a single line, a single point, the empty set, or the entire plane $\mathbb{R}^2$. In total seven different homeomorphism types occur among the sets in the family $(S_a)_{a \in \mathbb{R}^6}$.

(3.3) In Chapter 9 we prove more generally for o-minimal expansions of ordered rings that the sets of any given definable family belong to only finitely many definable homeomorphism types. (Two definable sets belong to the same definable

homeomorphism type if there is a definable homeomorphism between them.) This is not true for arbitrary o-minimal structures: all intervals of any given o-minimal structure belong to a single definable family, but given any cardinal κ there are o-minimal structures having κ mutually non-homeomorphic intervals, see Chapter 1, (6.6) and (7.11). Nevertheless, we will prove here some finiteness results on definable families in arbitrary o-minimal structures that are needed in the next chapters.

(3.4) In the following $\pi: R^{m+n} \rightarrow R^m$ denotes the projection on the first m coordinates.

(3.5) PROPOSITION.

- (i) Let C be a cell in R^{m+n} and $a \in \pi(C)$. Then C_a is a cell in R^n .
- (ii) Let $\mathcal{D}$ be a decomposition of R^{m+n} and $a \in R^m$. Then the collection

$$\mathcal{D}_a := \{C_a : C \in \mathcal{D}, a \in \pi(C)\}$$

is a decomposition of R^n .

PROOF. For $n = 1$ this is immediate from the definitions. Suppose the proposition holds for a certain n , and let C be a cell in $R^{m+(n+1)}$. Let $\pi_1: R^{m+(n+1)} \rightarrow R^{m+n}$ be the obvious projection map, so that $\pi \circ \pi_1: R^{m+(n+1)} \rightarrow R^m$ is the projection on the first m coordinates.

If $C = \Gamma(f)$, then $C_a = \Gamma(f_a)$, where $f_a: (\pi_1 C)_a \rightarrow R$ is defined by $f_a(x) = f(a, x)$. If $C = (f, g)_D$ with $D = \pi_1 C$, then $C_a = (f_a, g_a)_E$, where $E = D_a$.

In both cases it is clear that C_a is a cell in R^{n+1} . Property (ii) follows easily. $\square$

(3.6) COROLLARY. Let $S \subseteq R^m \times R^n$ be definable. Then there is a number $M_S \in \mathbb{N}$ such that for each $a \in R^m$ the set $S_a \subseteq R^n$ has a partition into at most M_S cells. In particular, each fiber S_a has at most M_S definably connected components.

PROOF. Take a decomposition $\mathcal{D}$ of R^{m+n} partitioning S . Then for each a in R^m the decomposition $\mathcal{D}_a = \{C_a : C \in \mathcal{D}, a \in \pi(C)\}$ of R^n consists of at most $|\mathcal{D}|$ cells and partitions S_a . So we can take $M_S = |\mathcal{D}|$. $\square$

The following consequence is worth recording.

(3.7) COROLLARY. Let $S \subseteq R^m \times R^n$ be definable. Then there is a natural number M_S such that for each $a \in R^m$ the set $S_a \subseteq R^n$ has at most M_S isolated points. In particular, each finite fiber S_a has cardinality at most M_S .

(3.8) The following exercise is intended for readers familiar with elementary logic.

EXERCISE.

Let $\mathcal{R} = (R, <, \dots)$ be an o-minimal L -structure and $\mathcal{R}' = (R', <', \dots)$ an L -structure elementarily equivalent to $\mathcal{R}$. Show that $\mathcal{R}'$ is also o-minimal.

Notes and comments

The monotonicity theorem for o-minimal structures on the real line has an easy proof, see Van den Dries [19]. The more general monotonicity theorem (1.2) is due to Pillay and Steinhorn [49]. The cell decomposition theorem for “strongly” o-minimal structures on the real line is also in [19]. The general cell decomposition theorem (2.11) is again more difficult and was established by Knight, Pillay and Steinhorn in [35]. It is worth keeping in mind that most proofs that a particular structure is o-minimal actually give at the same time the uniform finiteness property (2.13), that is, strong o-minimality. Thus we could have simplified this chapter considerably for those readers who want to restrict their attention to o-minimal structures on the real line for which the finiteness property (2.13) is available (as is usually the case). However, in the model-theoretic literature on o-minimal structures it is important to have the results available in the present generality, sometimes even in the sharper form given in (1.9), exercise 1, in (2.19), exercise 4, and in the exercise of (3.8).

